

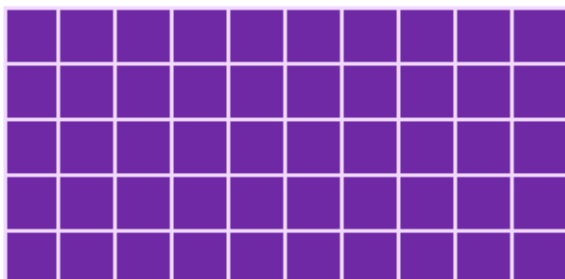
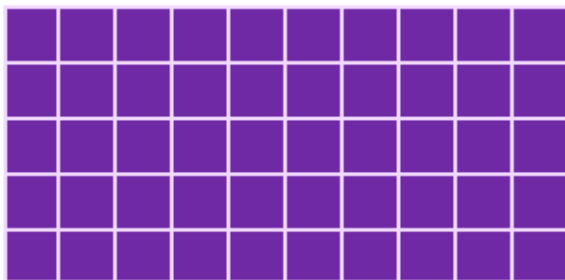
Lesson Objective

Analyze different rectangles and reason about their area.

Materials: Personal white board

1. Find the area of two equal rectangles by labeling side lengths and multiplying.

Label the side lengths of each rectangle. Then write a multiplication equation to find the area of one rectangle. What is the area of the other rectangle? What is the total area of the two rectangles?



Answer: 5 units by 10 units; $5 \times 10 = 50$, 50 square units; 50 square units; 100 square units

Point out that each rectangle is built on a unit grid. Have students count the unit squares along each side and label the side lengths as 5 units and 10 units.

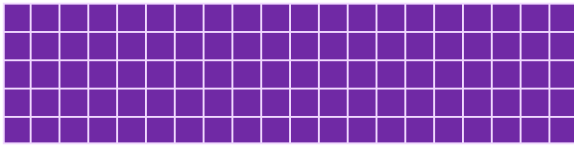
Ask students to write a multiplication equation for the area of one rectangle and solve. Confirm the area is 50 square units.

Ask students how they know the area of the second rectangle without multiplying again. Listen for the reasoning that the rectangles are equal, so the area is also 50 square units.

Have students add the two areas to find the total. Confirm $50 + 50 = 100$ square units is the total area of both rectangles together.

2. Find the area of a longer rectangle formed by joining two equal rectangles.

The two equal rectangles from Problem 1 are placed side by side to form one longer rectangle. Label the side lengths of the longer rectangle. Then find its total area.



Answer: 5 units by 20 units; 100 square units

Ask students how the new side lengths compare to the rectangles in Problem 1. Guide them to see that one side is still 5 units, and the longer side is $10 + 10$, or 20 units.

Ask students how they could find the total area. Prompt them to consider multiplying 5×20 or adding the two partial areas from Problem 1.

If students multiply, work through 5×20 as 5×2 tens, which is 10 tens, or 100 square units. If they add, confirm $50 + 50 = 100$ square units.

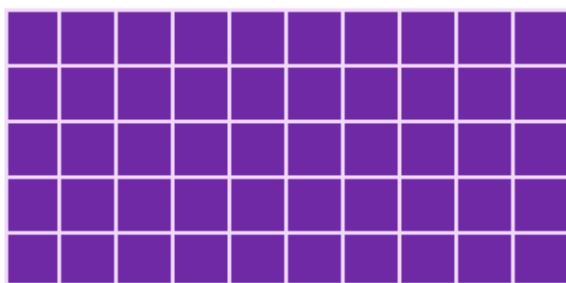
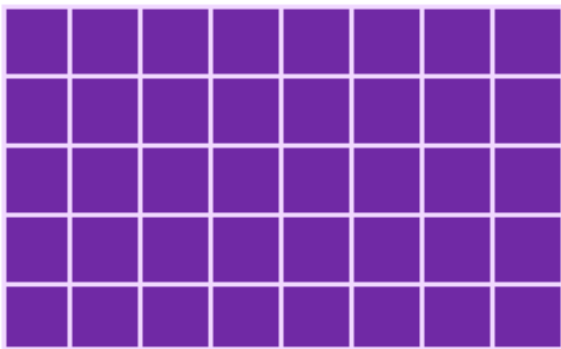
Point out that the total area of the longer rectangle is the same as the sum of the two smaller rectangles. The amount of square units does not change when the rectangles are pushed together.

3. Find the area of a longer rectangle formed by joining two unequal rectangles.

Two unequal rectangles are joined along a matching side to form one longer rectangle. Label the side lengths of the longer rectangle. Then find its total area.

Students work independently.

As students finish, ask them to explain how they found the area of the longer rectangle and how they knew the new long side was 18 units.



Answer: 5 units by 18 units; 90 square units

Monitor For

- Do students count the unit squares along each edge accurately to label the side lengths as 5 units, 8 units, and 10 units?
- Do students recognize that the two equal rectangles in Problem 1 have the same area without recomputing?

Instructional Moves

- If a student struggles to read a side length, have them count the unit squares along that edge of the rectangle and label each side before writing the equation.
- Reinforce the multiplication for area by having students say "5 rows of 10 square units is 5×10 , or 50 square units" as they write the equation.

• In Problem 2, do students see that the new long side is $10 + 10$, or 20 units, rather than counting all the squares one by one? • In Problem 3, do students add the two partial areas ($40 + 50$) rather than try to multiply 5×18 directly?	• For Problem 2, if a student is unsure how to multiply 5×20, prompt them to think of it as 5×2 tens, which is 10 tens, or 100. • For Problem 3, if a student tries to multiply 5×18, redirect them to find the area of each smaller rectangle first, then add: $40 + 50 = 90$ square units.
Reflection Questions • In Problem 1, how did you know the area of the second rectangle without multiplying again? <i>Answer: The two rectangles are equal, so they have the same area, 50 square units.</i> • In Problem 2, how did you find the long side of the new rectangle? <i>Answer: I added $10 + 10$ because the two equal rectangles were joined end to end.</i> • How did adding the areas of the two smaller rectangles in Problem 3 help you find the area of the longer rectangle? <i>Answer: The longer rectangle is made of the two smaller rectangles, so its area is the sum of their areas: $40 + 50 = 90$ square units.</i> • What stays the same when two rectangles are pushed together to form a longer rectangle? <i>Answer: The total number of square units stays the same.</i>	